

How do alveolar cells fail in lethal COVID-19?

An ablation-first, replication-audited trajectory analysis

[Author] — Systems Biology class presentation

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The question in one sentence

Do alveolar epithelial cells in lethal COVID-19 fail as a **coherent block** (pushed toward one damaged state), or as a **continuous graded collapse** (spread out in many directions)?

- “Block” $\Rightarrow$ expect **centroid displacement**.
- “Collapse” $\Rightarrow$ expect **dispersion / variance inflation**.
- These predict opposite things, so the failing test is informative.

Why this matters (and why I'm not claiming new biology)

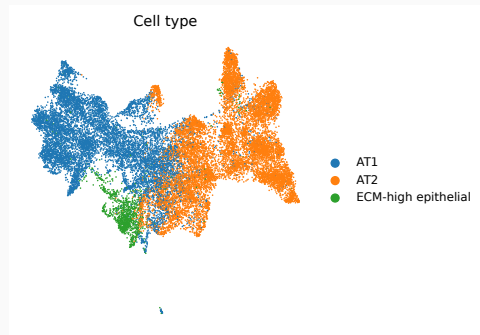
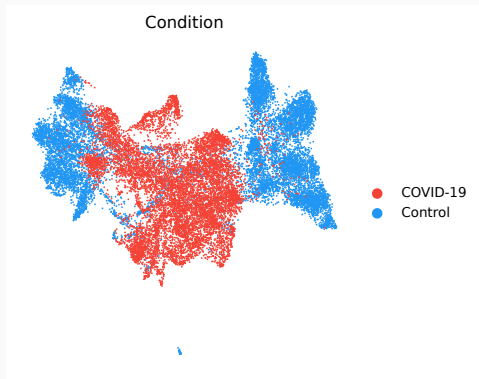
- Melms 2021, Delorey 2021, Bharat 2020 already described alveolar injury in COVID-19 lungs.
- My contribution is **methodological**: a pre-registered, ablation-first, replication-audited trajectory pipeline.
- Three pre-registered criteria, each with a null and a test statistic, declared *before* I looked at the numbers.

- **Primary:** SCP1219 (Melms 2021) — 116,313 snRNA-seq nuclei, 22,128 AT1/AT2/ECM-high, 7 healthy + 20 COVID-19 donors.
- **Replication:** cellxgene census (HLCA-extended + Krasnow), *SCP1219 removed by dataset_id*. 89,736 AT1/AT2 cells, 618 donors, 35 datasets.
- Primary has full transcriptome; replication has a curated 84-gene panel (enough to score programs, too narrow for a new manifold).

Pipeline at a glance

- Scanpy 1.12.1, harmonypy 0.2.0, Python 3.12.3, seed 42.
- Normalize $\rightarrow$ log1p $\rightarrow$ 3,000 HVGs $\rightarrow$ 50-PC PCA $\rightarrow$ Harmony on donor_id $\rightarrow$ kNN ($k=15$) $\rightarrow$ UMAP, diffusion map.
- Diffusion pseudotime rooted at the healthy-AT2 centroid, normalized to $[0, 1]$.
- 8 curated gene programs (`sc.tl.score_genes` vs `.raw`).
- 10 pre-specified ablations.

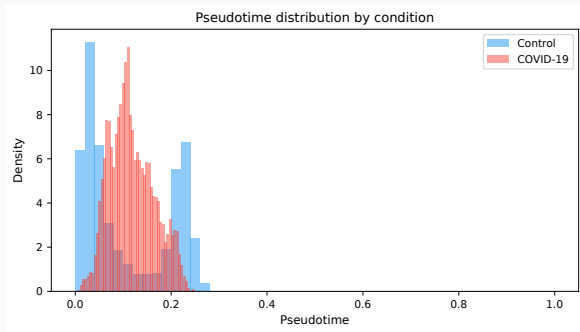
The manifold



Left: UMAP coloured by condition. Right: by cell type.

Result 1 — pseudotime shift (supported)

- Control median 0.063; COVID median 0.111; $\Delta = +0.0483$.
- 1,000-permutation null: $p = 0.001$.
- Leave-one-donor-out: direction preserved in **27/27** donors.



Result 2 — centroid displacement **fails**

- One-sided Mann–Whitney U in PCA-50 (COVID $>$ Healthy): $p = 1.0$.
- COVID median actually *lower* than healthy.

The “mono-directional damage” effect model is **wrong**. This is the most informative finding of the paper: the failure tells you the effect is *not* a pushed centroid.

Result 3 — dispersion (the real signal)

- Variance in PCA-50 distance to own centroid:
COVID **33.28** vs. Healthy **14.58** (2.3-fold).
- Levene statistic 276.5, $p < 10^{-30}$.

COVID-19 AT1/AT2 cells are *spread out, not pushed away*. This is the “robustness collapse” signature.

Ten ablations — the stress tests

Ablation	MPD	ρ	r	Notes	“Dispersion
1 AT2-only	+0.133	0.33	−0.29	sign flip	
2 UMAP / diffmap	+0.048	0.55	+0.14	unchanged	
3 Harmony corrected	+0.125	0.45	+0.14	amplifies shift	
3 Uncorrected	+0.048	0.55	+0.14	baseline	
4 No apoptosis genes	+0.074	0.50	+0.14	slight up	
5 4 roots	−0.037 to +0.048	−0.10 to 0.64	+0.14	root-dependent	
6 Broader epithelial	−0.008	0.83	+0.25	shift abolished	
7 LOO (27 donors)	0.028–0.202	varies	—	direction 27/27	
8 Curated-only	+0.048	1.00	+0.14	stable	
8 MSigDB-only	+0.048	−0.80	+0.14	ordering inverted	
9 Palantir	+0.028	—	—	alt. algorithm	
10 Permutation	$p = 0.001$	—	—	1000 shuffles	

and donor-level direction survive; fine ordering does not.”

The Harmony bug — worth a slide

- In v1, the corrected PCs were being written back with the wrong orientation (the PyTorch backend returns (n_cells, n_pcs) but the old CPU backend returned its transpose).
- Effect: “batch-corrected” branch was effectively noise.
- One-line orientation check fixed it. Post-fix:
 - MPD $+0.0483 \rightarrow +0.1246$ ($\approx 2.6\times$ **amplification**).
 - Dispersion and displacement claims unchanged.
- Moral: “batch correction didn’t help” can mean *the pipeline is broken*, not *the method doesn’t matter*.

Independent replication — dispersion

Metric	COVID	Normal
n cells	4,441	85,295
n donors	43	575
Injury composite variance	0.0505	0.0306
Variance ratio COVID/Normal		1.65 ×
Levene p	1.0×10^{-137}	

Dispersion replicates across 35

independent datasets. Effect size drops from $2.3\times$ to $1.65\times$ — expected with cross-study heterogeneity.

Replication — what works and what doesn't

- **Cell-level MW** (COVID > normal injury composite): $p = 1.0$ (dominated by a few large normal donors).
- **Donor-level pseudobulk MW**: $p = 1.98 \times 10^{-3}$ (replicates).
- **KRT8⁺/CLDN4⁺ DATP marker threshold**: COVID 22.9% vs Normal 47.6% (opposite direction!). Retracted as a portable cross-cohort metric — marker thresholds don't transfer, manifold-based states do.

Composite verdict

Criterion	Primary	Replication
Centroid displacement	fails ($p = 1.0$)	fails cell-level ($p = 1.0$)
Dispersion (Levene)	supported ($p \approx 0$)	supported ($p \approx 10^{-137}$)
Pseudotime enrichment	supported ($p = 0.001$)	supported donor-level ($p = 1.98\text{e-}3$)
Harmony amplifies	MPD $\times 2.6$	—
KRT8 ⁺ /CLDN4 ⁺	3.1 $\times$ enriched	0.48 $\times$ — retracted

The *robustness collapse* signature (dispersion + donor-level pseudotime) survives out-of-sample. The *directional centroid* and *marker-threshold DATP* signatures do not.

What I want you to take away

1. **Pre-register opposite-direction hypotheses.** A failing core test tells you the effect model is wrong, not just that n is small.
2. **Dispersion $>$ centroid** for heterogeneous disease states.
3. **Donor-level pseudobulk** beats cell-level tests in noisy multi-study atlases.
4. **Marker-threshold cell states don't transfer.** Use manifolds.
5. **Ablations catch bugs.** The Harmony fix here is exhibit A.

Limitations

- Primary cohort: post-mortem, lethal cases only, cross-sectional.
- Replication: Delorey (GSE171524) and Wendisch (PRJNA705687) are not on the census — full-transcriptome replication remains a gap.
- 84-gene replication panel is too narrow for an independent manifold.
- Fine program ordering is hypothesis-generating only (Ablation 8 MSigDB inverts it).

Thank you.
Questions?